protocol. Most application protocols in common use, such as HTTP, LDAP, or SMTP (email), all use TCP and so can use TLS to secure the connection. But a TCP implementation has some downsides when used in constrained IoT devices, such as the following:

- A TCP implementation is complex and requires a lot of code to implement correctly. This code takes up precious space on the device, reducing the amount of code available to implement other functions.
- TCP's reliability features require the sending device to buffer messages until they have been acknowledged by the receiver, which increases storage requirements. Many IoT sensors produce continuous streams of real-time data, for which it doesn't make sense to retransmit lost messages because more recent data will already have replaced it.
- A standard TCP header is at least 16 bytes long, which can add quite a lot of overhead to short messages.
- TCP is unable to use features such as *multicast* delivery that allow a single message to be sent to many devices at once. Multicast can be much more efficient than sending messages to each device individually.
- IoT devices often put themselves into sleep mode to preserve battery power when not in use. This causes TCP connections to terminate and requires an expensive TCP handshake to be performed to re-establish the connection when the device wakes. Alternatively, the device can periodically send keep-alive messages to keep the connection open, at the cost of increased battery and bandwidth usage.